SUPPLEMENTAL MATERIAL

Table S1. **Risk of bias assessed based on Newcastle-Ottawa assessment scale (NOS).**

Observational study	Selection	Comparability	Outcome
Diab et al., 2020	****	**	**
Saluan et al., 2018	**	*	***
Kume et al., 2018	***	**	***
Okita et al., 2016	****	**	***
Yoshioka et al., 204	****	**	***
Yeates et al., 2010	**	*	**
Eishi et al., 1995	***	*	**
Raman et al., 2016	**	*	**
Garcia-Cabrera et al., 2013	**	*	**

A study can be awarded a maximum of one star for each numbered item within the Selection (max 4*) and Outcome (max 3*) categories. A maximum of two stars can be given for Comparability.

Table S2. Demographic patient's characteristics of the nine observational studies included in the final analyses.

Observational study	Type of study	ICH patients (n)	Patients (n)		Definition of early/late surgery		Sex; n (m/f)	Age (mean ± SD) or median (IQR), years	Preoperative features				Indication for surgery		
			Early	Late	Early	Late			Valve involvement (%)	Large vegetation (> 10-15mm) in n (%)	Staphylococci-n (%)	EuroSCORE II (%); mean ± SD or median (IQR)	Heart Failure (NYHA class 3-4)	Uncontrolled infection	high embolic risk
Diab et al., 2020	single centre	34	21	11	≤7 d	8-14; 15-28, >28	m/f: 30/4	61 (47–71)	M-24%, A-35%, B-41%	21 (66 %)	15 (44 %)	25 (9–42)	19 (56 %)	7 (21%)	8 (24%)
Saluan et al., 2018	single centre	38	17	21	≤28 d	>28 d	-	-	-	-	-	-	-	-	-
Kume et al., 2018	single centre	25	17	8	<14 d	> 14 d	m/f: 20/5	56.8 ± 20.1	A-24%, M-76%, T-12%	-	-	11.8 ± 11.5	13 (31%)	-	-
Okita et al., 2016	multicentre	54	13	41	≤7 d	8-21; >21	m/f: 34/20	54.5 ± 17.4	A-37%, M-70%	26 (48%)	38 (32%)	-	26 (22%)	14 (12%)	70 (59%)
Yoshioka et al., 2014	single centre	30	5	25	≤7 d	8-14; 15-28, >28	m/f: 17/13	62 (51–63)	A-20%, M-66%, B-13%	9 (30%)	14 (46%)	-	-	-	-
Yeates et al., 2010	single centre	3	1	2	<14 d	> 14 d	-	-	A.25%, B-75%	-	-	-	-	-	-
Eishi et al., 1995	multicentre	34	2	32	not clear (<1, <14, <28d)		-	-	-	-	-	-	-	-	-
Raman et al., 2016	single centre	6	5	1	≤7 d	8-10 d	m/f: 5/1	-	M-66.7%, B-33.3%	5 (83%)	-	-	-	-	-
Garcia-Cabrera et al., 2013	multicentre	12	4	8	≤14 d	15-21; >21 d	-	-	-	19 (4%)	-	-	-	-	-

IQR: interquartile range; SD: standard deviation; m: male; f: female; A: aortic; M: mitral; T, tricuspid; B: both/multiple; NYHA: New York Heart Association.

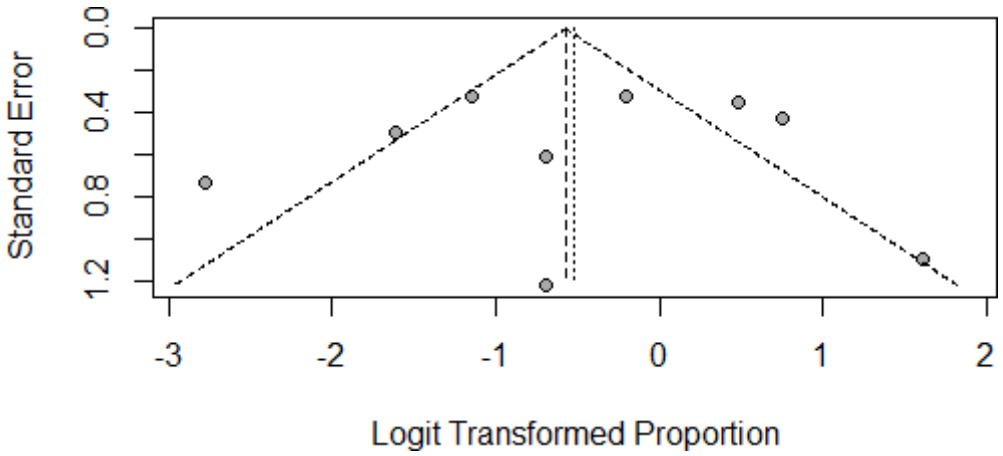
Table S3. Results of generalized linear mixed model with covariates early vs. late surgery and time point.

Characteristic	All-cause mortality		Neurological deterioration	
	OR (95% CI)[1]	**p-value**	**OR (95% CI)**[1]	**p-value**
Early				
Late	—		—	
Early	1.68 (0.95 to 2.98)	0.074	2.28 (1.18 to 4.39)	0.014
Time point:				
7 days	—		—	
14 days	1.08 (0.54 to 2.13)	0.83	0.94 (0.41 to 2.14)	0.89
21 days	0.97 (0.44 to 2.14)	0.93	0.86 (0.39 to 1.92)	0.72
28 days	0.90 (0.44 to 1.85)	0.77	0.72 (0.29 to 1.77)	0.47
Heterogeneity	Tau2= 0.69		Tau2= 2.97	
LRS early vs. late	0.26, df= 3, p = 0.96		0.61, df =3, p = 0.89	

OR = Odds Ratio; CI = Confidence Interval; Tau2: Heterogeneity variance; LRS = Likelihood Ratio Test

Figure S1. Publication bias tests: (A) Funnel plot; (B) Egger's linear regression test.

A



B

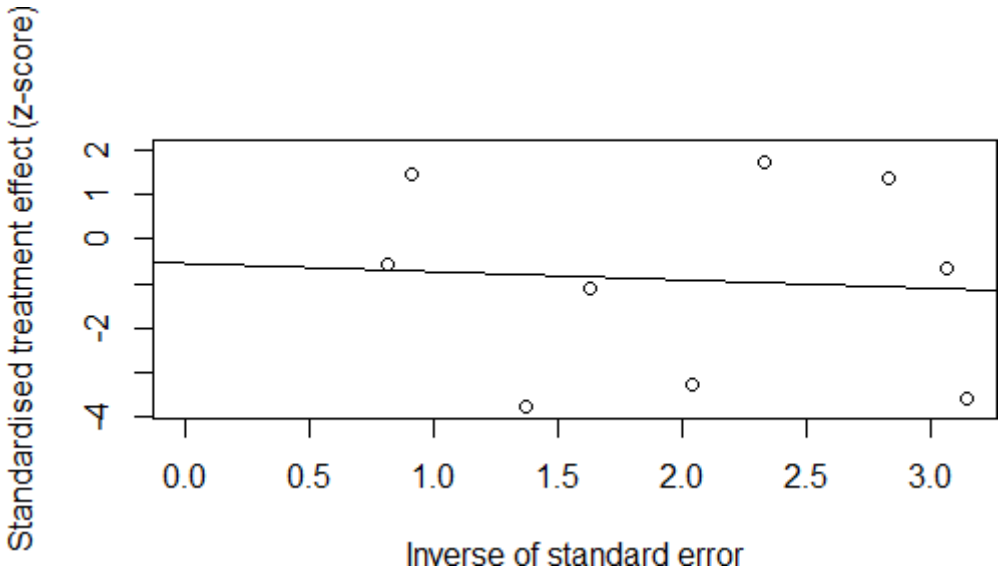


Figure S2. Kaplan-Meier Curve extracted from available studies in literature comparing the cumulative survival of IE-patients with an indication of surgery who were treated surgically vs. conservatively.

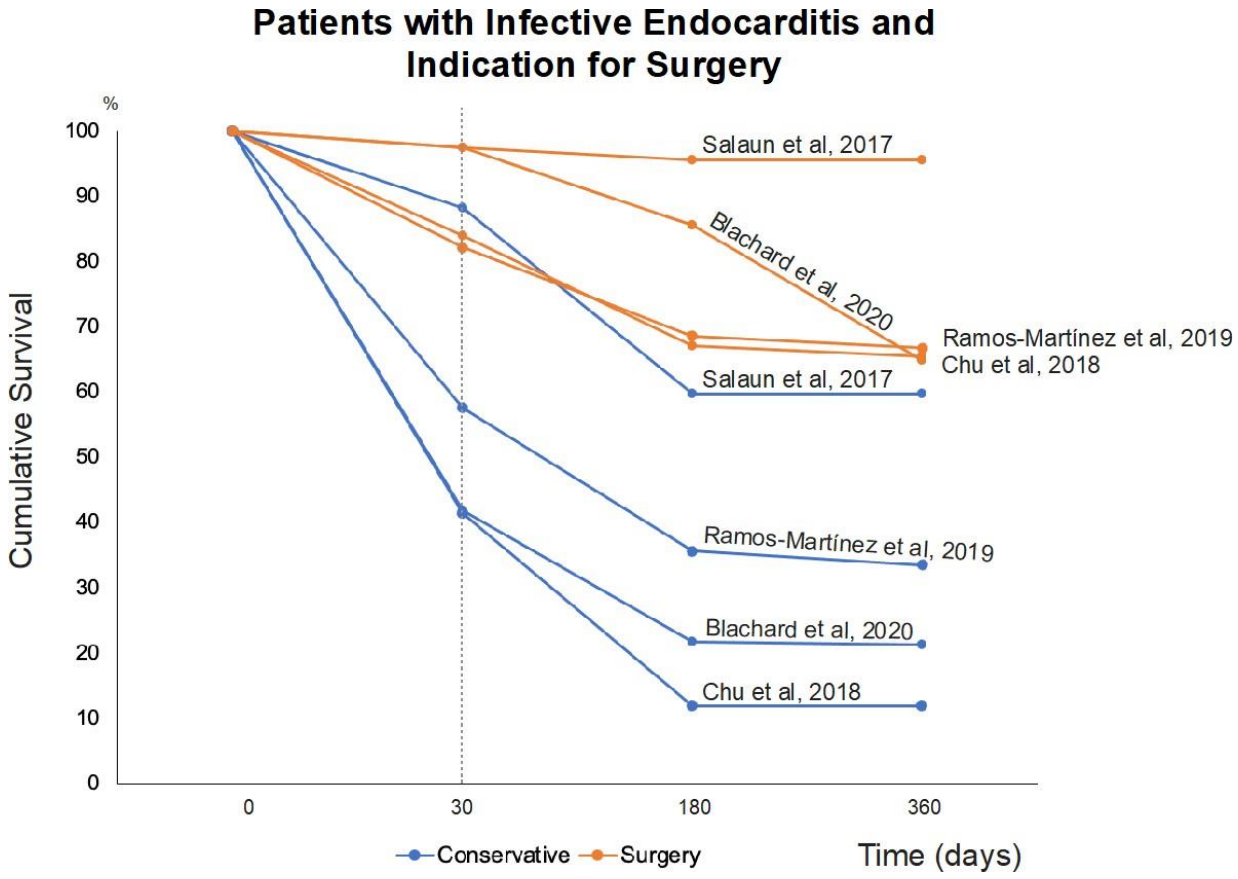


Figure S3. Forest plot of postoperative endpoint 1 (all-cause mortality within 30 days) in IE-patients with preoperative ICH who underwent early vs. late valve surgery with random slope and intercept model (RSM).

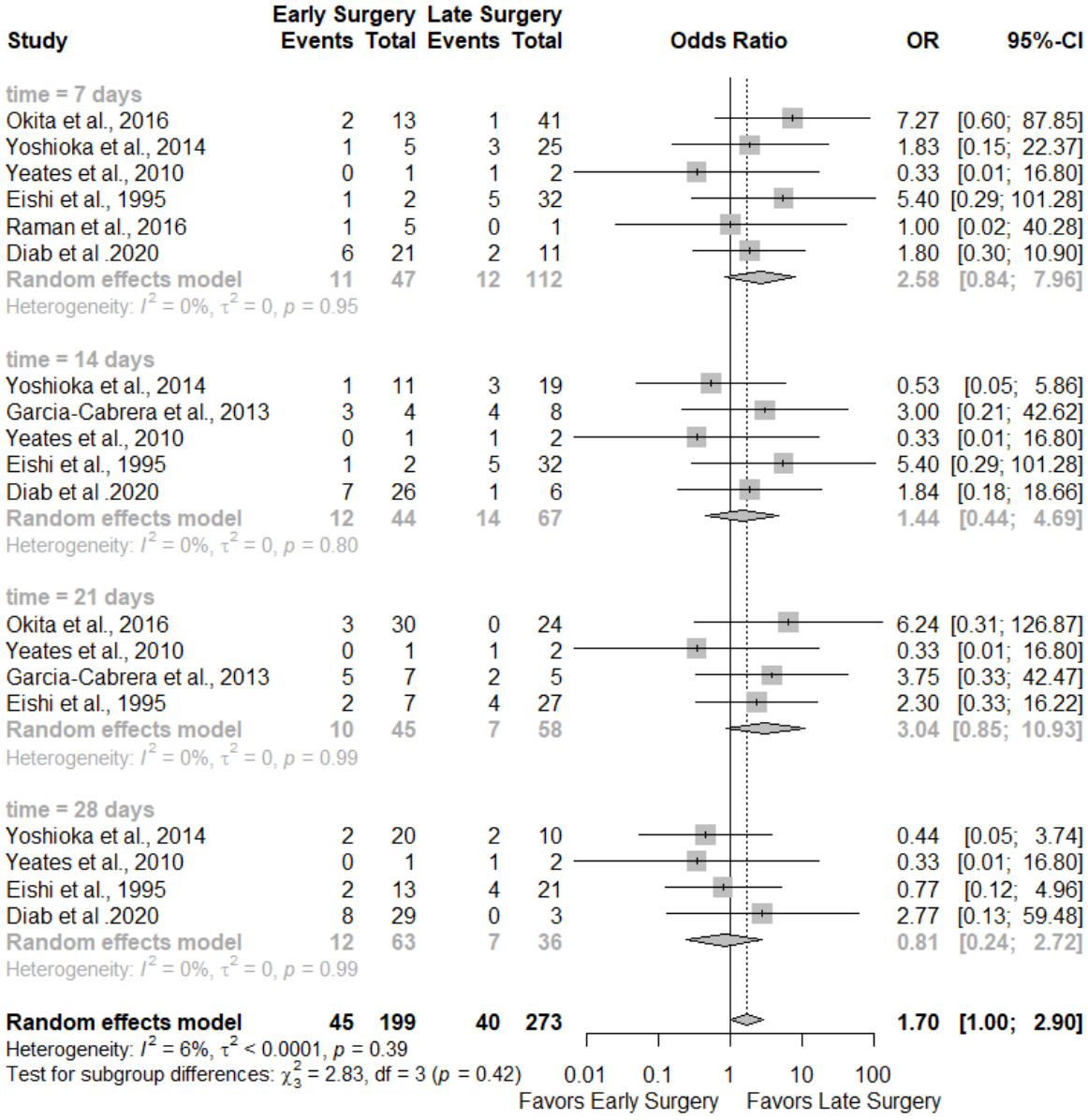


Figure S4. Forest plot of combined postoperative endpoint 2 (neurological deterioration) in IE-patients with preoperative ICH who underwent early vs. late valve surgery with random slope and intercept model (RSM).

